

deform a mesh. Each bone in the skeleton creates an envelope that affects the underlying vertices. Each vertex in the mesh is then weighted to each bone in the skeleton. A bone that completely affects a vertex will be given a weight of 1.0, while bones that do not affect the mesh are given a weight of 0.0. When multiple bones affect a vertex, their weights are somewhere between 1 and 0, but total weights always add up to 1.0. The Skin modifier also has tools to modify the envelopes as well as tools to assign weights manually and to paint weights. User-defined gizmos can be configured to affect the shape of the mesh depending on the position of a bone.

6.5.2 Modifying Weights

Vertex weights in 3ds Max can be modified in a number of ways. Envelopes surround each bone and provide the first level of weighting. Vertices that lie within the envelope are weighted to the corresponding bone. Weights can be configured manually or by painting, by using the tools in the Weight Properties rollout. The Skin Weight tool contains a palette of options to set weights for individual vertices or groups of vertices. Envelopes surround each bone and determine which vertices are affected. Envelopes can be moved and resized. Envelopes can be viewed by selecting the desired bone from the list in the Skin modifier. The Skin Weight table is a spread sheet like table that allows you to type in numeric weights for vertices while the Paint Weights tool allows you to paint vertex weights by using a brush.

6.5.3 Using Gizmos

Gizmos affect the shape of the mesh depending on the position of a bone. 3ds Max has three types of gizmos: Joint Angle, Bulge Angle, and Morph Angle. All of the gizmos are set up the same way. The affected bone and vertices are selected, and then the gizmo is added to the list. The joint is then animated through its range of motions, and the gizmos are modified to reshape the mesh according to the joint's angle. The key frames of the joint's animation are then deleted.

6.5.4 Skin Wrap Modifier

The Skin Wrap modifier is designed to let a low-resolution mesh deform a higher-resolution mesh. This works well for highly complex characters, because the actual mesh that's deformed is independent of the actual control mesh. This means you can modify the high resolution mesh and still keep the animation. The low-resolution object doing the deforming is called a control